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An Object-Based Approach Simulating Early Carbonate Diagenesis

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Abstract

The early diagenetic processes that occur in carbonate reservoirs play a pivotal role in shaping their porosity and permeability, ultimately influencing their suitability for hydrocarbon storage and CO₂ sequestration. As the foundation for subsequent mechanical and chemical compaction, early diagenesis sets the stage for the long-term evolution of these reservoirs. Therefore, accurately modeling these complex diagenetic processes is crucial for predicting the quality of carbonate reservoirs and optimizing assessments of their CO₂ storage capacity. To address this challenge, we have developed a groundbreaking object-based method, paired with an intuitive Graphical User Interface (GUI), designed to simulate the intricate processes of early carbonate diagenesis. This integrated approach situates diagenetic processes within a comprehensive stratigraphic and climatic framework, accounting for a range of key reactions that control the evolution of carbonate reservoirs. These reactions include mineral dissolution, cementation, precipitation, and stabilization, which are simulated across four distinct diagenetic zones: the vadose zone, meteoric phreatic zone, marine phreatic zone, and mixing zones. Characteristic of a carbonate ramp, these zones are critical in determining the overall diagenetic trajectory of the reservoir. To ensure the accuracy of our simulations, reaction rates are calculated using advanced reactive transport modeling techniques, which are rigorously validated against empirical field data. By leveraging this innovative approach, we aim to provide a more nuanced understanding of the complex interplay between diagenetic processes, stratigraphy, and climate, ultimately enhancing our ability to predict and optimize the quality of carbonate reservoirs for a range of applications, including CO₂ storage and hydrocarbon production.

Introduction

Early diagenesis is a critical stage in the evolution of sediments, encompassing the initial chemical, physical, and biological transformations that occur shortly after deposition, when the sediments are still in an unconsolidated state. This complex process takes place at relatively low temperatures and pressures, typically within the uppermost sections of the sediment column, where the material is saturated with water. The early diagenetic environment is characterized by a dynamic interplay of factors, including the decay of organic matter, the precipitation of minerals from pore waters, and changes in the texture and structure of the sediments. These transformations can have a profound impact on the sediment's physical and chemical

properties, influencing its porosity, permeability, and reactivity. The study of early diagenesis is essential for understanding the formation of sedimentary rocks, as it provides valuable insights into the processes that control the preservation of fossils, the generation and migration of hydrocarbons, and the development of economic deposits of minerals. Furthermore, the knowledge of early diagenetic processes has significant practical implications for various industries, including agriculture, water management, and the exploration for mineral resources and hydrocarbons. By elucidating the complex interactions between sediments, water, and organisms during the early diagenetic stage, researchers can provide valuable inputs for a range of applications, from environmental management and resource exploration to climate change mitigation and geological hazard assessment.

Reactive Transport Models (RTMs) are sophisticated computational tools that simulate the intricate interplay between chemical reactions and physical transport processes in porous media, such as soil, rock, and water systems. These models are particularly crucial in the oil and gas industry, where they are used to understand and predict the complex processes of early diagenesis, which can significantly impact the quality and behavior of hydrocarbon reservoirs (e.g., [Lu et al. 2023](#)). RTMs typically involve the solution of complex mathematical equations, including those that describe advective and diffusive transport, chemical reactions, and thermodynamic equilibria, which require significant computational power and advanced numerical methods to solve ([Xu et al., 2012](#)). The application of RTMs can span a wide range of scales, from the pore-scale interactions that control chemical reactions and transport at the microscopic level, to large watershed or basin-scale processes that influence the behavior of entire ecosystems ([Steeffel, 2005](#)). These models are essential for understanding and predicting the behavior of contaminants, nutrients, and other substances in natural and engineered environments. The versatility and predictive power of RTMs make them essential tools for understanding and managing a wide range of environmental and industrial systems, including groundwater flow and quality, soil remediation, geological carbon sequestration, and the optimization of hydrocarbon recovery.

Object-based modeling (OBM) is a powerful and flexible approach to simulating complex systems and processes in a wide range of fields, including geology, ecology, engineering, and economics. At its core, OBM involves representing a system or process as a collection of discrete objects, each with its own set of attributes, properties, and behaviors. These objects can interact with one another and their environment in complex ways, allowing for the simulation of emergent phenomena and the exploration of "what-if" scenarios. In contrast to traditional grid-based or continuum-based modeling approaches, OBM allows for a more nuanced and realistic representation of complex systems, where individual components or entities can have unique characteristics and behaviors. This can be particularly useful in fields such as geology, where the behavior of individual rocks, sediments, or minerals can have a significant impact on the overall system. Object-based models can be used to simulate a wide range of processes, from the movement of individual particles or objects to the evolution of entire ecosystems or landscapes. They can also be used to integrate multiple types of data and models, allowing for a more comprehensive and interdisciplinary understanding of complex systems.

This study presents the development of a novel object-based early diagenesis modeling engine, designed to simulate the complex chemical reactions that occur in a carbonate ramp setting. This engine utilizes a series of quantified reaction rates to predict the chemical transformations that take place during early diagenesis, without the need for explicit RTM simulations. The model incorporates a range of key chemical reactions that are critical to the early diagenetic process, including the dissolution of minerals, cementation, precipitation, and stabilization. These reactions are simulated in a variety of environmental settings, encompassing both arid and humid climates, as well as different marine chemistries, such as the Calcite Sea and Aragonite Sea. By accounting for the complex interplay between these chemical reactions and the prevailing environmental conditions, the model provides a robust and flexible framework for predicting the early diagenetic evolution of carbonate ramps. The model's ability to simulate early diagenesis in a range of climatic and marine settings makes it a valuable tool for understanding the controls on carbonate

reservoir quality and for predicting the potential for hydrocarbon accumulation in these systems. Overall, this study demonstrates the potential of object-based modeling for simulating complex geological processes, such as early diagenesis, and highlights the advantages of this approach for improving our understanding of carbonate reservoir systems.

Methodology

Prior to outlining our approach, we will introduce several relevant terms.

Diagenetic environments

Early diagenesis is a critical stage in the formation of sedimentary rocks, occurring at or near the surface, or in the shallow subsurface, where the sediment is still in contact with the overlying water column or atmospheric conditions. In this environment, the sediment is subject to a range of diagenetic processes that can significantly alter its composition, texture, and structure. According to Longman (1980), four major diagenetic environments are typically encountered in the shallow subsurface, each characterized by distinct fluid chemistry and pore fluid distribution patterns (Fig. 1). These environments include:

1. Marine phreatic environments, where the sediment is fully saturated with seawater, leading to the formation of marine cements and the preservation of primary textures and structures.
2. Meteoric vadose environments, where the sediment is exposed to freshwater from precipitation, runoff, or groundwater, resulting in the formation of vadose cements and the alteration of primary minerals.
3. Meteoric phreatic environments, where the sediment is saturated with freshwater, leading to the formation of freshwater cements and the dissolution of primary minerals.
4. Mixing environments, where the sediment is exposed to a mixture of marine and freshwater, resulting in the formation of complex cements and the alteration of primary textures and structures.

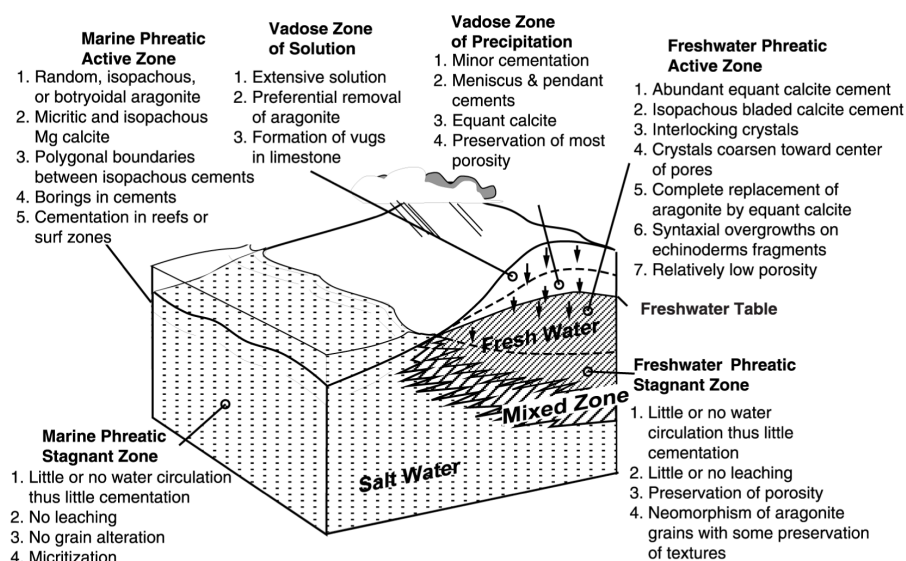


Figure 1—Summary of early diagenetic processes and products that occur in carbonate environments. https://wiki.aapg.org/Early_carbonate_diagenesis#cite_note-ch09r35-1.

Each of these diagenetic environments is characterized by unique cement textures and alteration features, which can provide valuable information about the conditions under which the sediment was formed and altered. However, the interpretation of diagenetic sequences can be challenging, as a rock may be exposed to multiple environments over its history, resulting in a complex overprint of diagenetic features. For example,

a rock may initially form in a marine phreatic environment, only to be later exposed to meteoric vadose conditions, followed by a period of burial and compaction. This complex history can make it difficult to unravel the sequence of diagenetic events and to reconstruct the conditions under which the rock formed.

Calcite Sea vs. Aragonite Sea

The composition of seawater during the Phanerozoic Eon, which spans the past 541 million years, has undergone significant changes, with profound implications for the Earth's geochemical cycles and the formation of sedimentary rocks. One of the most notable changes is the oscillation in the primary inorganic marine calcium carbonate mineralogy, which has alternated between calcite (calcite sea) and aragonite sea) over geological time scales (Hardie, 1996; Fig. 2). This shift in mineralogy has been driven by secular variations in the major ion chemistry of seawater, which have been confirmed by the analysis of fluid inclusions in marine halite (Lowenstein et al., 2001; 2003; Horita et al., 2002). The observed variations in seawater chemistry are thought to be controlled primarily by fluctuations in the mid-ocean ridge hydrothermal brine flux, which is driven by changes in the rate of ocean crust production (Hardie, 1996). This, in turn, is influenced by the Earth's tectonic cycles, including the formation and breakup of supercontinents, the creation of oceanic crust, and the resulting changes in global ocean chemistry. The interplay between these geological processes has resulted in a complex and dynamic history of seawater chemistry.

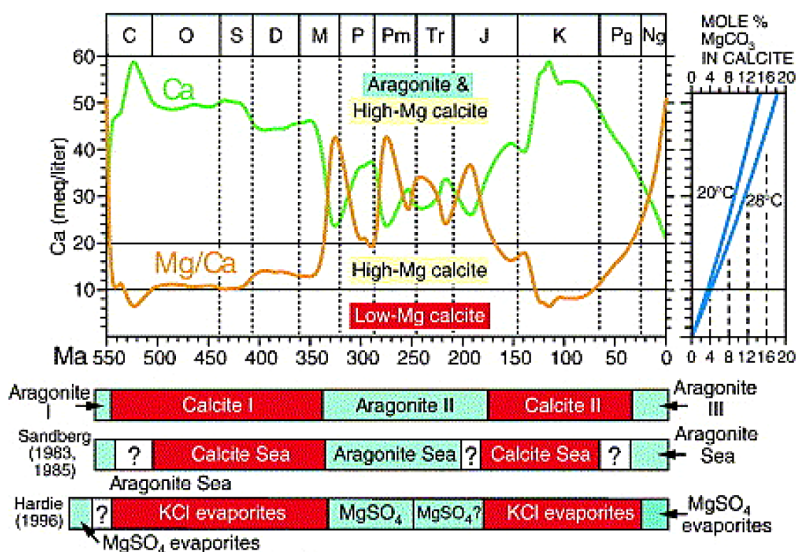


Figure 2—Secular variation of seawater chemistry in Ca, Ca/Mg and mineralogy (Sandberg 1983).

Mineral Stabilization

Aragonite → calcite; HMC → calcite

Calcium carbonate (CaCO_3) is a versatile mineral that exists in three distinct polymorphs: vaterite, aragonite, and calcite. The formation of these polymorphs is highly dependent on various environmental factors, including water chemistry, pH, water-to- CO_2 ratios, agitation rates, and temperature. The specific conditions under which CaCO_3 precipitates can lead to the formation of one of these three polymorphs. Water chemistry plays a crucial role in determining the polymorph formed, with factors such as $\text{Ca}^{2+}:\text{CO}_3^{2-}$ ratios, ionic strength, and the presence of inhibiting ions like magnesium, sulfate, manganese, strontium, phosphate, and organic matter influencing the precipitation process (Morse et al., 2007). Vaterite, the amorphous form of calcium carbonate, is the least stable of the three polymorphs. Under normal geological conditions, vaterite rapidly transforms into calcite, a process that can occur over a period of months to years

(Addadi et al., 2003; De Leeuw & Parker, 1998; Nancollas & Sawada, 1982). Due to its transient nature, vaterite is not considered a significant component in our modeling efforts.

Calcite, on the other hand, is a more stable polymorph that can exist in various forms, depending on its magnesium content. Low-magnesium calcite (LMC) contains less than 4 mol% MgCO_3 , while high-magnesium calcite (HMC) contains more than 8 mol% MgCO_3 . High-Mg calcite crystals with magnesium contents exceeding 10 mol% are thermodynamically unstable under ambient conditions (Raz et al., 2000). However, high-magnesium calcites with magnesium contents ranging from 4 to 45 mol% are commonly found in many marine organisms, such as foraminifera, echinoids, red corals, red corallina algae, and calcareous serpulid worms, in modern seas (Long et al., 2014). The presence of magnesium ions appears to be essential for the formation of a transient amorphous calcium carbonate phase, which then incorporates magnesium into its structure, resulting in Mg-containing calcite (Raz et al., 2003). Notably, high-Mg calcite is even less stable than aragonite, which is already a metastable phase under normal geological conditions. The instability of high-Mg calcite is attributed to its high magnesium content, which can lead to increased solubility and reactivity. This instability has significant implications for the formation and preservation of calcium carbonate minerals in various geological and biological settings. Understanding the complex relationships between these polymorphs and the factors that influence their formation is crucial for predicting the behavior of calcium carbonate in different environments and for interpreting the geological record.

Zonal early diagenesis

To implement early diagenesis, different zones of the carbonate ramp model (Fig. 1) are treated differently.

Marine Phreatic Zone. The marine-phreatic diagenetic environment is characterized by the complete saturation of pore space in a sediment or rock with normal marine water. Despite the fact that seawater is typically supersaturated with respect to calcium carbonate, the precipitation of carbonate cement from a single pore volume of water is often limited to a very small amount (Matthews, 1974). Significant cementation requires a substantial flux of water, which is not typically observed in the interior of platforms. As a result, the marine phreatic environment can be divided into two distinct zones: (1) the active marine phreatic zone, where water movement, in combination with other processes, leads to cementation, and (2) the stagnant marine phreatic zone, where water movement through the sediment is relatively slow, or other factors, such as fluid chemistry, inhibit cementation, resulting in little to no cementation.

In the stagnant marine phreatic zone, micritization, which is the process of boring algae and fungi breaking down the sediment, is a significant diagenetic process. This process can occur at depths of over a meter beneath the sediment-water interface (May and Perkins, 1979), and is not limited to the active marine phreatic zone, although it may also occur in this environment. Micritization can lead to the creation of microporosity, which refers to the formation of small pores within the sediment, but may not necessarily affect the bulk porosity of the sediment. However, in this version of the model, the change in microporosity resulting from micritization is not considered. Instead, the focus is on the larger-scale diagenetic processes that control the overall porosity and permeability of the sediment. The distinction between the active and stagnant marine phreatic zones is important, as it highlights the complex interplay between water movement, fluid chemistry, and biological processes that control the diagenetic evolution of sediments in marine environments.

Meteoric Vadose Zone. The vadose diagenetic environment refers to the subaerial zone that lies below the land surface and above the water table, where the pore space in the sediment or rock is not fully saturated with water. This zone, also known as the unsaturated zone, extends from the surface soil to the groundwater table and is characterized by the presence of both air and water in the pore space. The vadose zone is a dynamic environment that is influenced by a range of factors, including rainfall, plant roots, and microbiota activity. The water that passes through this zone can drive mineral dissolution, particularly when it is acidic,

making it undersaturated with respect to high-magnesium calcite (HMC), aragonite, and calcite. The acidity of the water is often controlled by the atmospheric $p\text{CO}_2$, with present-day rainwater having a pH of around 5.5. Additionally, plant root decay and microbiota activity can generate organic acids, which can further contribute to the dissolution of minerals. As the water flows through the carbonate sediments, it can quickly reach equilibrium, leading to significant dissolution and the generation of secondary porosity. The amount of water (flux) that passes through the vadose zone is a critical control on the extent of dissolution and porosity generation, which is why the diagenetic profiles in humid and arid climates can be quite different.

However, dissolution is not the only diagenetic process that occurs in the vadose zone. When the vadose water reaches saturation, it can begin to precipitate calcite through evaporation or CO_2 degassing (Harris, 1971). This process can lead to the formation of new minerals and the reduction of porosity. The vadose zone can be conveniently divided into two parts, which represent the end members of a continuous spectrum: the zone of dissolution and the zone of precipitation, also known as the capillary fringe zone. The zone of dissolution is characterized by the dissolution of minerals, while the zone of precipitation is characterized by the precipitation of new minerals. The size and extent of these zones can vary greatly depending on local conditions, such as climate, vegetation, and the thickness of the vadose zone. In some cases, the zone of precipitation may be missing entirely. The boundary between the zones is gradational and can fluctuate significantly over short periods of time, particularly in response to rainfall events. Climate plays a crucial role in controlling the diagenetic processes that occur in the vadose zone, with factors such as temperature, humidity, and precipitation rate influencing the extent of dissolution and precipitation. As a result, the diagenetic profiles in different climates can be quite distinct, reflecting the unique combination of factors that control the diagenetic processes in each environment.

Zone of dissolution: The zone of dissolution is a critical component of the vadose diagenetic environment, where the interaction between meteoric water and limestone leads to the dissolution of calcium carbonate. This zone is typically located at the top of the vadose zone, but its extent can vary greatly depending on the depth to the water table and the rate at which the meteoric water becomes saturated with respect to calcium carbonate. In some cases, the zone of dissolution can extend downward for tens or even hundreds of meters, resulting in significant dissolution of the limestone and the formation of characteristic karst topography. This type of landscape is commonly found in humid climates, where the high levels of rainfall and groundwater flow lead to the development of extensive cave systems and other solutional features. Caves, in particular, tend to form at the top of the freshwater phreatic zone, where saturated vadose waters may become cooled or mixed with waters in equilibrium with a lower $p\text{CO}_2$, leading to the precipitation of calcium carbonate and the formation of speleothems (Thraillkill, 1968). On the other hand, degassing of the waters can cause them to become supersaturated with respect to calcium carbonate, leading to the precipitation of calcium carbonate near the water table.

In more moderate climates, the solution of homogeneous limestone can produce a range of distinctive features, including pitted outcrop surfaces and vugs. These features are formed through the dissolution of the limestone by meteoric water, which can create a network of small holes and cavities within the rock. The only cementation that occurs in this zone is in the form of evaporitic and/or biologic caliche crusts, which form at or near the air/rock interface. These crusts are composed of calcium carbonate that has been precipitated from solution, often through the evaporation of water or the activity of microorganisms. They can provide valuable information about the diagenetic history of the rock and the conditions under which it formed. Overall, the zone of dissolution plays a critical role in shaping the morphology and geochemistry of limestone rocks in the vadose diagenetic environment, and its characteristics can vary greatly depending on the specific climate, hydrology, and geological setting.

Zone of precipitation: The zone of precipitation is a critical component of the vadose diagenetic environment, where cementation occurs through the precipitation of minerals from solution. Although cementation in the vadose zone is a relatively minor process compared to dissolution, it can still play

a significant role in shaping the morphology and geochemistry of the rock. The zone of precipitation is typically located below the zone of solution, in the lower parts of the vadose zone, where the water table or a perched water table is present. The amount of cementation that occurs in this zone is dependent on several factors, including the degree of saturation, the factors causing precipitation, the flow rate, and time.

The degree of saturation is a critical control on cementation, as it determines the amount of water available for precipitation. When the water is fully saturated with respect to a particular mineral, precipitation can occur, leading to the formation of cements. The factors causing precipitation can include changes in temperature, pressure, or chemistry, such as the loss of CO₂ from the system, which can cause a shift in the equilibrium between the water and the minerals. The flow rate of the water also plays a crucial role, as it controls the amount of water from which material is being withdrawn for precipitation. A slower flow rate can lead to more extensive precipitation, as the water has more time to interact with the rock and precipitate minerals. Time is also an important factor, as cementation is a gradual process that occurs over a prolonged period.

In the zone of precipitation, the precipitation of minerals can occur through a variety of mechanisms, including evaporation, CO₂ degassing, and temperature changes. For example, as water evaporates, the concentration of minerals in the solution increases, leading to precipitation. Similarly, as CO₂ is lost from the system, the pH of the water increases, causing the precipitation of minerals such as calcite. The resulting cements can include a range of minerals, such as calcite, aragonite, and dolomite, which can provide valuable information about the diagenetic history of the rock and the conditions under which it formed. Overall, the zone of precipitation plays a critical role in shaping the morphology and geochemistry of rocks in the vadose diagenetic environment, and its characteristics can vary greatly depending on the specific geological setting and hydrological conditions.

Meteoric Phreatic Zone. The meteoric phreatic zone is a critical component of the diagenetic environment, situated between the vadose zone and the mixed marine phreatic-freshwater zone. This zone is characterized by full saturation with fresh water, which enters the zone through the vadose zone or directly through streams, lakes, and other surface water bodies. The phreatic zone is notable for its high diagenetic efficiency, which is attributed to the presence of a large volume of water that facilitates the interaction between the water and the rock. According to [Harris \(1971\)](#), a significant proportion of the water in the coastal phreatic zone of Barbados, up to 99%, enters the zone through streams, lakes, and other runoff from the interior of the island, rather than passing directly through the soil zone or vadose zone. The size and geometry of the meteoric phreatic zone are influenced by a range of factors, including rainfall amounts, pore type, relief, and the distribution of porosity and permeability in the rocks.

As water enters the meteoric phreatic zone from the vadose zone, it may be strongly undersaturated with respect to calcium carbonate (CaCO₃) initially. However, as it moves downward through the freshwater phreatic zone, it will become increasingly saturated with CaCO₃. This leads to a zonation of diagenetic processes within the zone, with distinct areas characterized by different types of mineral dissolution and precipitation. At the top of the zone, there is an area where both calcite and aragonite are dissolved, due to the undersaturated nature of the water. Beneath this zone, there is a region where only aragonite dissolves, as the water becomes more saturated with respect to calcite. Further down, there is a zone of penecontemporaneous aragonite solution and calcite precipitation, where the dissolution of aragonite is accompanied by the precipitation of calcite. Still lower, there are zones of active calcite cementation and no solution, where the water is fully saturated with respect to calcite and aragonite, and cementation occurs through the precipitation of calcite. Finally, there is a stagnant zone of little water movement, where diagenetic processes are slowed due to the lack of fluid flow. This zonation of diagenetic processes within the meteoric phreatic zone highlights the complex interactions between the water, rock, and minerals, and demonstrates the importance of considering the hydrological and geochemical factors that control diagenesis in this environment.

Mixing Zone. The boundary between the marine phreatic and freshwater phreatic environments is a critical zone of interaction, where the mixing of waters from both sources creates a brackish water environment. This zone, often referred to as the mixing zone, is characterized by a complex interplay of hydrological and geochemical processes that can significantly alter the initial porosity distribution of the rocks. The freshwater recharge and seawater circulation in this zone can lead to the creation of a fluid that is undersaturated with respect to calcite, causing significant dissolution and cave formation. This process is particularly pronounced near the coast, where the flux of both freshwater and seawater is high, resulting in a zone of intense diagenetic activity. The dissolution of calcite in this zone can lead to the formation of caves, sinkholes, and other karst features, which can have a profound impact on the porosity and permeability of the rocks.

The cements produced in the mixing zone are relatively rare, as the undersaturated nature of the fluid tends to favor dissolution over precipitation. However, it's worth noting that the mixing of meteoric and marine waters can create a fluid that remains saturated with respect to dolomite, even if it's undersaturated with respect to calcite. While dolomitization can occur in the mixing zone, it's not a common mechanism for dolomite formation and is often limited to rare situations and locations. As a result, dolomitization in the mixing zone is not considered in this module, and the focus is instead on the more prevalent processes of calcite dissolution and precipitation. Nevertheless, the mixing zone remains an important area of study, as it provides valuable insights into the complex interactions between freshwater and seawater, and the resulting diagenetic processes that shape the porosity and permeability of rocks in coastal environments.

Results and discussion

The methodology is then implemented with MATLAB app designer and a series of 200 kyrs long simulations of Calcite Sea (as an example in this work) under Humid and Arid climates are conducted, showing the minerals and porosities changing with depositional time for comparison.

Figure 3 illustrates the diagenetic environments (zones) for a Calcite Sea carbonate ramp model in humid and arid climates, comprising four distinct zones: Meteoric Vadose, Meteoric Phreatic, Mixing Zone, and Marine Phreatic. An early diagenesis process is applied to this zonal model. The primary difference between the two climate scenarios is the thickness of the mixing zone, which in turn affects the size of the Meteoric Phreatic and Marine Phreatic zones.

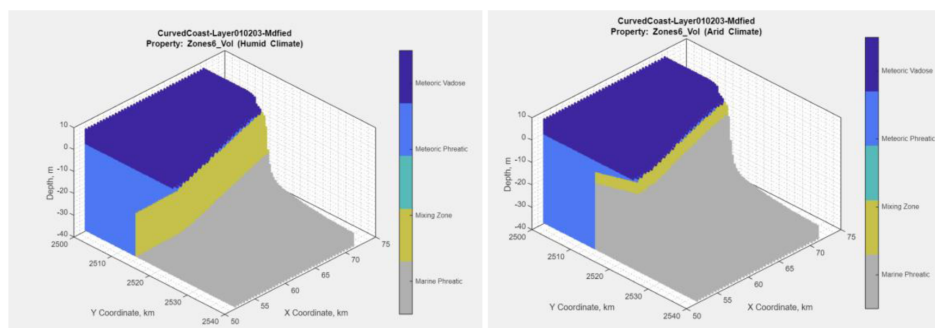


Figure 3—Display of diagenetic zones for a Calcite Sea carbonate ramp model in Humid (left) and Arid (right) climates.

Figure 4 illustrates the changes in Aragonite mineral percentage over time, specifically after 0 (present day), 10 kyrs, and 200 kyrs of diagenesis, for a Calcite Sea model under both Humid and Arid climate conditions. The results are presented in two layers, with the top layer representing the Humid climate scenario and the bottom layer representing the Arid climate scenario. A comparison of these results reveals that the total amount of Aragonite decreases significantly over the 200 kyrs period, from a high of 5% to a low of 1%, particularly in the meteoric phreatic and mixing zones near the water table in the Humid climate.

In contrast, the decrease in Aragonite is relatively less pronounced in the Arid climate scenario, especially in the same zones.

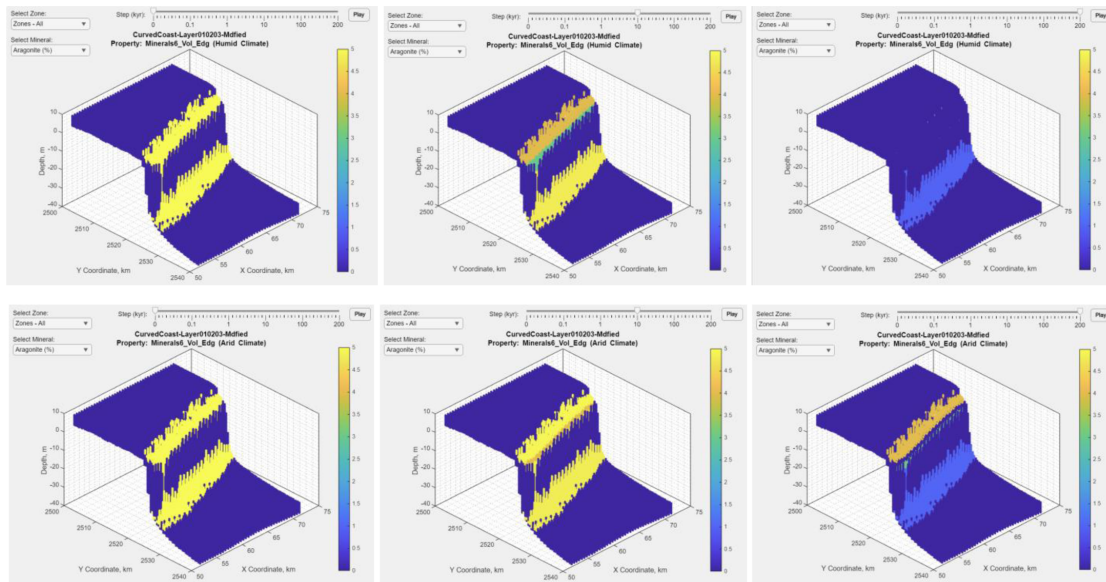


Figure 4—Display of simulated Aragonite changing after 0 (left), 10 kyrs (mid.) to 200 kyrs (right) diagenesis for a Calcite Sea in both Humid (top) and Arid (bottom) climates.

Figure 5 shows the changes in High-Magnesium Calcite (HMC) mineral percentage over time, specifically after 0 (present day), 10 kyrs, and 200 kyrs of diagenesis, for a Calcite Sea model under both Humid and Arid climate conditions. A comparison of these results reveals that, in the Humid climate scenario, the total amount of HMC decreases significantly over the 200 kyrs period, from 20% to 0% in the mixing zone, and from 10% to 0% in the Meteoric Vadose and Marine Phreatic zones. In contrast, the decrease in HMC is relatively slower in the Arid climate scenario, with a less pronounced reduction in the same zones over the same time period.

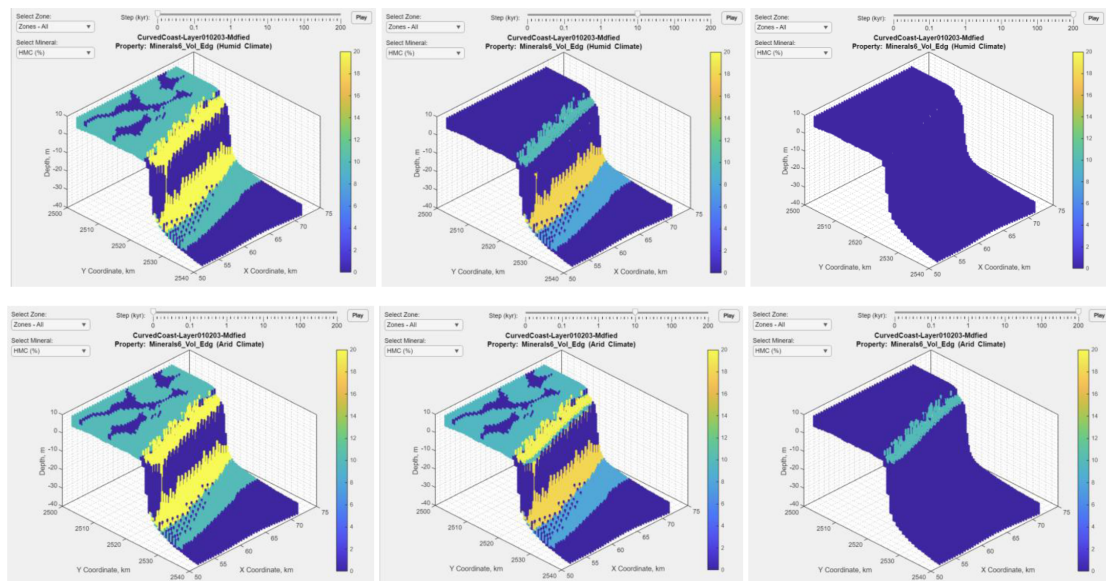


Figure 5—Display of simulated HMC changing after 0 (left), 10 kyrs (mid.) to 200 kyrs (right) diagenesis for a Calcite Sea in both Humid (top) and Arid (bottom) climates.

Figure 6 illustrates the changes in Calcite mineral percentage over time, specifically after 0 (present day), 100 kyrs, and 200 kyrs of diagenesis, for a Calcite Sea model under both Humid and Arid climate conditions. A comparison of these results reveals that, in the Humid climate scenario, the total amount of Calcite decreases significantly over the 200 kyrs period, from 90% to 70% in the Meteoric Vadose zone, and from 70% to 45% in the Mixing zone. In contrast, the decrease in Calcite is relatively slower in the Arid climate scenario, with a less pronounced reduction in the same zones over the same time period.

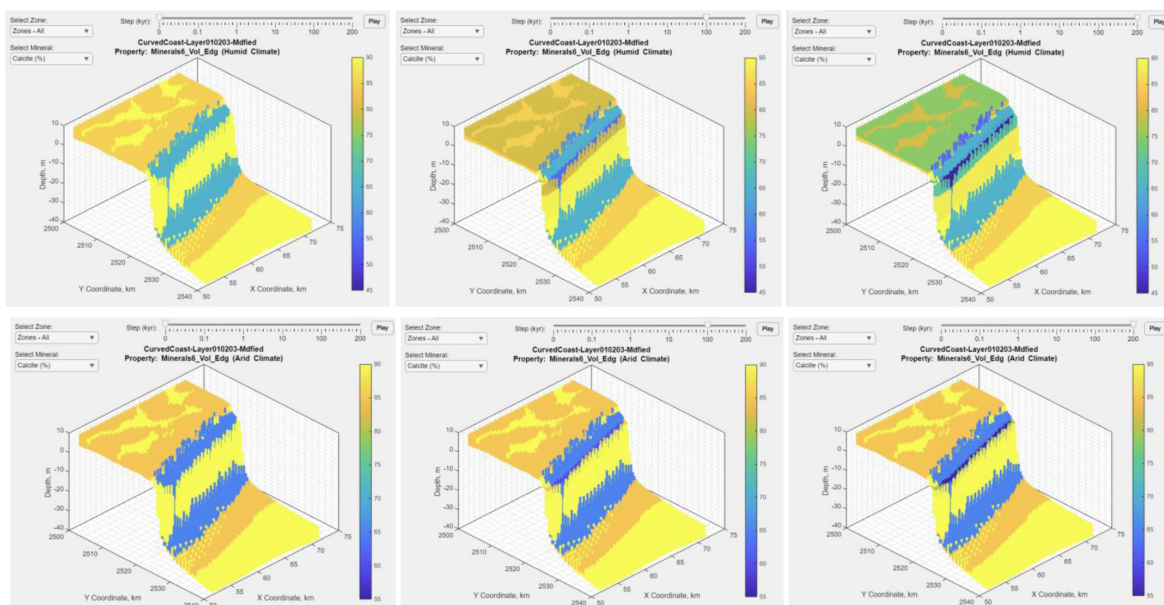


Figure 6—Display of simulated Calcite changing after 0 (left), 100 kyrs (mid.) to 200 kyrs (right) diagenesis for a Calcite Sea in both Humid (top) and Arid (bott.) climates.

Figure 7 illustrates the changes in formulated Calcite Cement percentage over time, specifically after 0 (present day), 100 kyrs, and 200 kyrs of diagenesis, for a Calcite Sea model under both Humid and Arid climate conditions. A comparison of these results reveals that, in the Humid climate scenario, the total amount of Calcite Cement increases substantially over the 200 kyrs period, reaching levels of 25% or higher in the Mixing zone and the upper part of the Marine Phreatic zone, and up to 10% in the lower Meteoric Vadose zone and the middle part of the Marine Phreatic zone. In contrast, the increase in Calcite Cement is relatively slower in the Arid climate scenario, with less pronounced growth in the aforementioned zones, but surprisingly, a more significant increase is observed in the upper Meteoric Vadose zone.

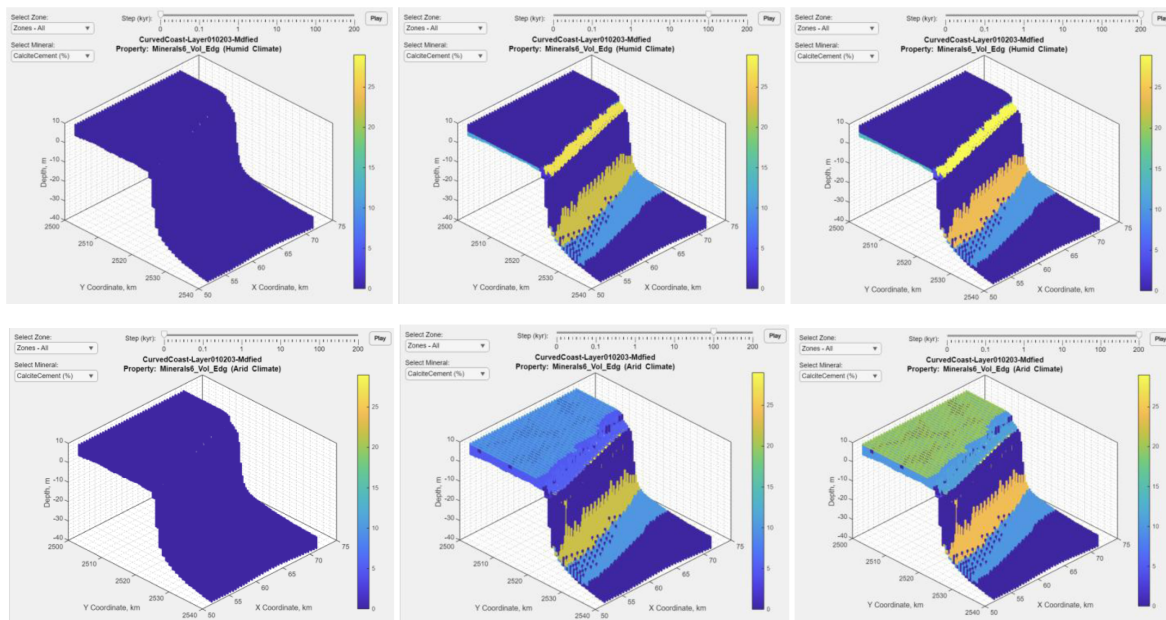


Figure 7—Display of simulated CalciteCement changing after 0 (left), 10 kyrs (mid.) to 200 kyrs (right) diagenesis for a Calcite Sea in both Humid (up) and Arid (down) climates.

Figure 8 illustrates the changes in calculated Porosity percentage over time, specifically after 0 (present day), 100 kyrs, and 200 kyrs of diagenesis, for a Calcite Sea model under both Humid and Arid climate conditions. A comparison of these results reveals that, in the Humid climate scenario, the total porosity increases significantly over the 200 kyrs period, with values rising from an average of 50% to over 60% in the Mixing zone and the upper part of the Meteoric Vadose zone. Notably, the upper part of the Meteoric Vadose zone exhibits the most pronounced increase, with porosity values reaching as high as 90%. In contrast, the increase in porosity is relatively less pronounced in the Arid climate scenario, with smaller areas of increase observed in the upper Meteoric Vadose and Mixing zones.

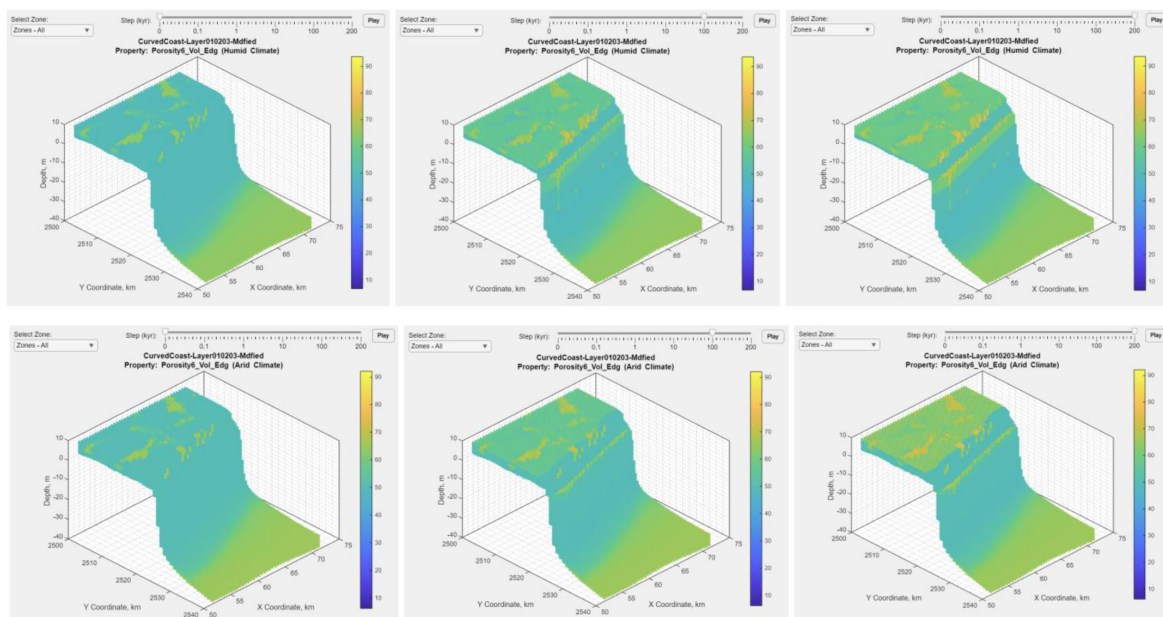


Figure 8—Display of simulated Porosity changing after 0 (left), 100 kyrs (mid.) to 200 kyrs (right) diagenesis for a Calcite Sea in both Humid (top) and Arid (bottom) climates.

The simulation of an Aragonite Sea in both Humid and Arid climates follows a similar analysis process to that of the Calcite Sea, and therefore, the details of this analysis are not repeated here.

Conclusions

Our findings reveal that diagenetic reactions reach their peak intensity in the meteoric phreatic and mixing zones, followed by the meteoric vadose and marine phreatic zones. This spatial distribution of diagenetic activity is consistent with the expected geochemical gradients and fluid flow patterns in these zones. Notably, humid climates exhibit pronounced dissolution patterns near the water table in meteoric phreatic and mixing zones, leading to significant changes in mineral composition and porosity. In contrast, arid conditions favor the formation of cementation layers or caliche crusts in the meteoric vadose zone, resulting in distinct mineralogical and petrophysical signatures. Among the minerals studied, high-magnesium calcite proved most prone to alteration, with aragonite also exhibiting significant reactivity. In contrast, calcite demonstrated relative stability, while quartz showed negligible reactivity due to its low solubility and inertness. Moreover, sedimentary deposits originating from aragonite seas exhibited greater mineral stabilization compared to those derived from calcite seas, highlighting the importance of considering the initial mineralogical composition in diagenetic modeling.

The success of our model lies in its ability to replicate the spatiotemporal distributions of mineral reactions, product formations, and porosity fluctuations with reliability and speed. By incorporating quantifiable rates of comprehensive diagenetic reactions into a zonation-based framework, our approach provides a robust and flexible tool for simulating early carbonate diagenesis. Furthermore, the accompanying graphical user interface (GUI) offers an intuitive platform for users to apply early diagenesis modeling to diverse carbonate systems, including those with complex mineralogical and geochemical characteristics. The GUI enables researchers to efficiently explore various carbonate scenarios, visualize temporal trends, and analyze diagenetic histories in detail, making it an invaluable resource for understanding the complex processes that shape carbonate reservoirs and aquifers. This study introduces a novel approach in early carbonate diagenesis modeling, one that has the potential to revolutionize our understanding of these critical geological systems and inform more effective strategies for their exploration, development, and management.

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